

Subject2Image: Evaluating Adaptation Approaches for Subject-Driven Image Generation

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Abstract

Subject-driven image generation enables text-to-image personalization from a small set of reference photos. In this work, we systematically evaluate subject-driven personalization across three distinct pipelines, comparing Low-Rank Adaptation (LoRA) and Textual Inversion. We benchmark all implementations on the standard DreamBooth dataset using a comprehensive suite of quantitative metrics: CLIP-T for prompt alignment, CLIP-I and DINO-I for subject identity preservation, and LPIPS for perceptual diversity. Experimental results demonstrate that LoRA-based weight adaptation significantly outperforms Textual Inversion strategies, with the first LoRA pipeline achieving the highest overall performance in subject fidelity and visual preservation.

1 Introduction

Subject-driven image generation aims to synthesize novel images of a specific target subject across arbitrary contexts using only a few reference photos. Leveraging large-scale pretraining, modern diffusion models - such as Stable Diffusion and FLUX - demonstrate remarkable open-domain synthesis capabilities. Personalization techniques, including Textual Inversion and DreamBooth, extend these foundation models by adapting them to specific concepts or visual identities.

In this work, we present a systematic empirical evaluation of subject-driven personalization using the modern Stable Diffusion 3.5 architecture. Specifically, we investigate how parameter-efficient fine-tuning strategies (via LoRA) and textual embedding optimization (via Textual Inversion) perform, both on the standard DreamBooth benchmark dataset.

To comprehensively explore the spectrum of subject-driven personalization, we implement and compare three distinct adaptation pipelines:

- **LoRA Baseline:** A parameter-efficient fine-tuning setup that integrates Low-Rank Adaptation into the DreamBooth framework to update the transformer weights.

- **LoRA Custom:** An extended variant of the first pipeline, combining LoRA-based weight adaptation with spatial data augmentation on the fine-tuning reference set.
- **Textual Inversion Custom:** A modern adaptation of Textual Inversion scaled to the Stable Diffusion 3.5 architecture, optimizing text embeddings within its multi-encoder continuous prompt space.

Additionally, an unpersonalized generation baseline is included in our quantitative and qualitative evaluation as an anchor to measure the base model’s intrinsic text-prompt alignment without subject adaptation.

To rigorously assess the three pipelines, we perform a comprehensive evaluation across a suite of complementary quantitative metrics. We measure text-prompt alignment using **CLIP-T**, subject identity preservation through **CLIP-I** and **DINO-I** and perceptual diversity via **LPIPS**. Through this multi-pipeline comparative analysis, we evaluate the trade-offs between weight-space tuning versus text-space optimization, and the efficacy of spatial data augmentation.

2 Related Work

2.1 Diffusion Models for Image Synthesis

Diffusion models have emerged as a powerful paradigm for high-quality image generation. Denoising Diffusion Probabilistic Models (DDPMs) formulate generation as the reverse of a gradual noising process, where a neural network learns to recover clean samples from progressively corrupted observations [1]. This formulation provides a stable and expressive framework for generative modeling and has subsequently become the foundation of many text-to-image generation systems.

More recently, the architecture underlying diffusion models has evolved beyond conventional convolutional U-Nets. Diffusion Transformers (DiTs) replace the U-Net backbone with a Transformer architecture, demonstrating that Transformer-based models can effectively scale diffusion-based image synthesis [2]. This architectural shift is particularly relevant for large-scale generative models, as it enables the use of highly scalable Transformer architectures.

Beyond architectural modifications, recent research has focused on refining the underlying generative dynamics. Specifically, Esser et al. [3] explore Rectified Flow formulations within Transformer architectures, demonstrating that modeling continuous-time trajectories along straight paths improves scaling behavior and sample quality at high resolutions. Collectively, these developments reflect a broader trend toward combining continuous-time flow dynamics with scalable, attention-based backbones.

2.2 Personalization of Text-to-Image Models

Despite their strong generative capabilities, general-purpose text-to-image models struggle to preserve specific subject identities across prompts. Personalization techniques address this limitation by adapting pretrained models to a few reference images of a target subject.

Early approaches like Textual Inversion [4] operate entirely in the text-embedding space while keeping model weights frozen. By optimizing a single pseudo-word vector to represent the visual concept, it enables lightweight personalization without altering the underlying network parameters. Conversely, DreamBooth [5] fine-tunes the generative model itself. By binding the subject to a unique text identifier and employing a class-preservation loss alongside subject reconstruction, it avoids language drift and captures high-fidelity identity features, even if at a significantly higher computational cost.

To bridge the gap between full model fine-tuning and embedding-only optimization, Low-Rank Adaptation (LoRA) [6] provides a Parameter-Efficient Fine-Tuning (PEFT) alternative. By freezing base weights and injecting trainable low-rank decomposition matrices into attention layers, LoRA drastically reduces parameter footprint and memory overhead while preserving generative capacity.

Overall, these foundational approaches navigate the trade-off between adaptation capacity and computational efficiency. Textual Inversion [4] optimizes embedding vectors, DreamBooth [5] fine-tunes full backbones, and LoRA [6] offers a parameter-efficient compromise. Together, these strategies establish the principles for modern subject-driven generation in flow- and Transformer-based architectures.

2.3 Data Augmentation in Subject-Driven Generation

Subject-driven personalization typically relies on extremely limited reference datasets - often consisting of only 3 to 5 images - making models highly prone to overfitting to static background cues, framing, or specific orientations. To encourage robust identity preservation, classical spatial transformations are frequently integrated into fine-tuning workflows [7]. Explicit geometric operations, such as random cropping, rotation, and horizontal flipping, prevent attention mechanisms from memorizing rigid spatial configurations, forcing the network to isolate invariant, fine-grained subject features rather than surrounding context.

3 Method

3.1 Problem Formulation and Base Architecture

Subject-driven personalization aims to adapt a pretrained text-to-image model to a specific target subject using a small set of reference images $\mathcal{X} = \{x_i\}_{i=1}^N$. The subject is bound to a unique textual identifier `<sks>` within a text prompt y (e.g., *“a photo of a <sks> dog”*).

Our framework leverages Stable Diffusion 3.5, a continuous-time generative model based on a Multimodal Diffusion Transformer (MM-DiT) backbone trained via a Rectified Flow formulation. Rather than predicting noise, the network f_θ learns a vector field pointing from noise distribution $x \sim \mathcal{N}(0, \mathbf{I})$ to clean image distribution $z \sim p_{\text{data}}$ along a linear trajectory $x_t = (1 - t)x + tz$. The training trajectory is optimized using the velocity matching objective:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t \sim U(0,1), x, z, c} [\|f_\theta(x_t, t, c) - (z - x)\|_2^2], \quad (1)$$

where $c = \tau_{\text{text}}(y)$ represents the text embedding generated by the language encoders.

3.2 LoRA Baseline: Parameter-Efficient DreamBooth

To adapt Stable Diffusion 3.5 without updating its full parameter set, our baseline pipeline integrates Low-Rank Adaptation (LoRA) within the DreamBooth framework. We freeze the base model parameters $W_0 \in \mathbb{R}^{d \times k}$ across the MM-DiT attention projections and inject trainable low-rank decomposition matrices $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$, with rank $r \ll \min(d, k)$. The forward pass for a linear projection layer is modified as:

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} B A x, \quad (2)$$

where α is a constant scaling hyperparameter.

During fine-tuning, only parameters $\Theta_{\text{LoRA}} = \{A, B\}$ are updated via equation (1), significantly reducing the computational overhead while preventing catastrophic forgetting.

3.3 LoRA Custom: PEFT with Data Augmentation

With the idea of mitigating overfitting to the limited reference set $\mathcal{X}$, our second pipeline extends the LoRA baseline by introducing an explicit image-space data augmentation operator $\mathcal{T}$ during optimization. Rather than modifying the original reference set $\mathcal{X}_{\text{orig}} = \{x_i\}_{i=1}^{N_{\text{orig}}}$ (with N_{orig} ranging from 4 to 6), we construct an additional set of augmented samples $\mathcal{X}_{\text{aug}}$ to reach a predefined target dataset size N_{target} .

Specifically, these additional samples are generated by sequentially replicating the original images in their initial order and applying a stochastic spatial transformation operator $\mathcal{T}$:

$$\tilde{x}_{N_{\text{orig}}+i} = \mathcal{T}(x_i) = (T_{\text{crop}} \circ T_{\text{flip}} \circ T_{\text{rot}})(x_i), \quad \text{for } i = 1, \dots, N_{\text{target}} - N_{\text{orig}}, \quad (3)$$

where T_{flip} applies random horizontal flipping with probability p_{flip} , T_{crop} performs random cropping, and T_{rot} applies a small random rotation.

The final augmented dataset $\mathcal{X} = \mathcal{X}_{\text{orig}} \cup \mathcal{X}_{\text{aug}}$ (such that $|\mathcal{X}| = N_{\text{target}}$) is then used during fine-tuning. Each image $x \in \mathcal{X}$ is mapped to its latent representation $z = \mathcal{E}(x)$ via the VAE encoder to optimize the velocity matching loss $\mathcal{L}_{\text{CFM}}(\Theta_{\text{LoRA}})$. By combining pristine reference photos with spatially perturbed copies, this strategy encourages the attention mechanisms to learn robust identity invariants rather than memorizing static composition boundaries.

3.4 Textual Inversion Custom: Multi-Encoder TI

Our third pipeline adapts Textual Inversion to the modern Stable Diffusion 3.5 architecture. Although SD 3.5 employs a triple language encoder system - allowing text-space personalization to theoretically optimize a tuple of three continuous embeddings $\mathbf{v} = (v_1, v_2, v_3)$ - we selectively optimize a dual embedding tuple $\mathbf{v} = (v_1, v_2)$ across two text encoders.

While keeping the MM-DiT backbone f_θ and all text encoders frozen, the embedding tuple $\mathbf{v}$ is optimized on the unaugmented reference set $\mathcal{X}_{\text{orig}}$ using the Flow Matching velocity objective (1), where $c = \tau_{\text{text}}(y; \mathbf{v})$ projects the learned embeddings into their corresponding textual feature spaces before concatenation.

This design choice significantly reduces execution time while maintaining target performance, as empirical observations indicated that optimizing all three encoders yielded no perceptible visual improvement in output quality.

4 Experiments and Results

4.1 Experimental Setup

Dataset. We evaluate our approach on the standard DreamBooth dataset, consisting in a set of 30 subjects encompassing both rigid objects and complex living subjects. For each subject, we utilize N_{orig} reference images (with N_{orig} ranging from 4 to 6) for training and an evaluation suite of diverse text prompts testing contextual adaptation and subject preservation.

Implementation Details. Our evaluation encompasses three distinct personalization pipelines, all based on the Stable Diffusion 3.5 model.

All experiments are executed on a single NVIDIA GPU.

LoRA Setup. For parameter-efficient fine-tuning (*LoRA Baseline* and *LoRA Custom*), LoRA is injected into the query, key, value, and output projections of both the standard and additional attention pathways of SD 3.5 with rank $r = 32$ and scaling factor $\alpha = 32$.

In *LoRA Custom* pipeline, spatial data augmentation expands the dataset to a target size N_{target} , which consists in 10 images. The augmentation transformation resizes input images by a factor of 1.15 (to 1177×1177 pixels), applies a random rotation in the range $[-20^\circ, 20^\circ]$, and extracts a random crop back to the target resolution of 1024×1024 pixels, alongside a random horizontal flipping ($p_{\text{flip}} = 0.5$).

Fine-tuning is conducted for 1200 steps using the AdamW 8-bit optimizer at an initial learning rate of 1×10^{-4} , regulated by a Learning Rate Scheduler.

Unlike standard DreamBooth setups, our LoRA fine-tuning omits the prior preservation loss, optimizing the network using solely the standard Conditional Flow Matching loss function (1).

Textual Inversion Setup. Textual Inversion Custom optimizes a double-embedding tuple across SD 3.5’s text encoders, both trained for 1500 steps with frozen model backbones.

Embedding optimization is conducted using the AdamW 8-bit optimizer at a fixed learning rate of 1×10^{-3} .

Inference Setup. During inference, images are generated using 28 denoising steps with a Classifier-Free Guidance (CFG) scale of 4.5.

Evaluation Metrics. We evaluate performance across four quantitative metrics, alongside a joint fidelity score:

- **CLIP-T:** Measures text-prompt alignment by computing cosine similarity between prompt text embeddings and generated image embeddings. Following standard practice in personalization benchmarks, scores are scaled by a factor of 2.5 to bring the metric values closer to a maximum bound of 1.0.
- **CLIP-I:** Evaluates high-level visual subject similarity by computing cosine similarity between CLIP image embeddings of the reference images and generated outputs. However, CLIP is less effective than DINO at preserving fine-grained identity, as it focuses on broad semantic alignment rather than instance-level details.
- **CLIP-T $\times$ CLIP-I:** Computes the product of scaled CLIP-T and CLIP-I to evaluate the joint trade-off between prompt alignment and subject identity preservation.
- **DINO-I:** Assesses fine-grained subject identity preservation using ViT-based DINO features, which are particularly sensitive to instance-level structural and textural details.
- **LPIPS:** Quantifies perceptual diversity among generated samples for the same text prompt using deep feature distance.

4.2 Quantitative Results

Table 1 reports the quantitative evaluation across all four configurations, comparing the three subject personalization pipelines alongside an unpersonalized generation baseline.

Table 1: Comparison between the three pipelines and an unpersonalized image generation pipeline. Best results are shown in **bold**.

Pipeline	CLIP-T $\uparrow$	CLIP-I $\uparrow$	CLIP-T $\times$ CLIP-I $\uparrow$	DINO-I $\uparrow$	LPIPS $\downarrow$
Generation Only	0.8953	0.6100	0.5461	0.2846	0.7919
LoRA Baseline	0.8678	0.6509	0.5648	0.6303	0.7134
LoRA Custom	0.8723	0.6451	0.5627	0.6252	0.7201
TI Custom	0.8763	0.6175	0.5411	0.3016	0.7780

As reported in Table 1, the fine-tuning approaches exhibit clear trade-offs between text adherence and subject identity preservation. The unpersonalized baseline achieves the highest text alignment (CLIP-T = 0.8953), as it is completely unconstrained by subject retention, but fails on instance identity metrics (DINO-I = 0.2846).

Overall, parameter-efficient fine-tuning via LoRA outperforms embedding-based optimization (Textual Inversion) across subject preservation metrics. Among the LoRA variants, the unaugmented baseline achieves the highest subject fidelity, obtaining top scores in CLIP-I (0.6509), CLIP-T $\times$ CLIP-I (0.5648), and DINO-I (0.6303), while also demonstrating lower LPIPS diversity.

Regarding text-space optimization, Textual Inversion yields quantitative metrics that closely mirror the unpersonalized **Generation Only** baseline. Nevertheless, **TI Custom** exhibits a marginal increase in subject fidelity over standard generation (DINO-I = 0.3016 vs. 0.2846 and CLIP-I = 0.6175 vs. 0.6100), indicating that embedding optimization still imparts a subtle conditioning effect, albeit far short of full identity preservation.

4.3 Qualitative Results

Figure 1 presents a visual comparison of generated samples across the three evaluated personalization pipelines and the unpersonalized one.



Figure 1: Comparison between the three pipelines and an unpersonalized image generation pipeline.

Visual inspection confirms the quantitative metric trends. Comparing fine-tuning paradigms, both LoRA-based pipelines achieve substantially higher subject fidelity than the Textual Inversion variant, successfully capturing intricate visual details, unique surface textures, and exact object geometry across novel background contexts. Conversely, Textual Inversion methods struggle to preserve fine-grained instance identity, often producing generic category representations or displaying semantic drift toward prompt background concepts.

Among the parameter-efficient fine-tuning approaches, **LoRA Baseline** and **LoRA Custom** yield broadly comparable context adaptation. Nevertheless, the unaugmented **LoRA Baseline** exhibits a consistent edge in accurately preserving the structural fidelity of the target subject, as **LoRA Custom** tends to repeat some details.

Finally, outputs from Textual Inversion are often visually indistinguishable from the **Generation Only** baseline. While the pipeline reliably adheres to prompt instructions and scene contexts - benefiting from the unconstrained generative model - it largely fails to inject the target identity, typically producing generic, class-level representations rather than preserving the unique visual signature of the reference subject. Subtle identity preservation effects remain discernible only across a select few subjects, reflecting the minor quantitative gains previously observed in section 4.2.

5 Discussion

Our experimental findings reveal a clear performance hierarchy among the evaluated personalization approaches:

$$\text{LoRA Baseline} > \text{LoRA Custom} \gg \text{TI Custom (SD 3.5)}$$

Below, we analyze the architectural and optimization dynamics that account for these comparative trade-offs.

Weight-Space Adaptation vs. Text-Space Optimization. The substantial performance gap between LoRA-based fine-tuning and Textual Inversion highlights the fundamental capacity limit of embedding-only personalization. Fine-tuning enables the network to explicitly capture complex 3D geometry, high-frequency textures, and distinct visual features. In contrast, embedding optimization yields to very low subject-identity preservation, capturing only a general idea of the subject.

Effect of Spatial Data Augmentation. Comparing **LoRA Baseline** and **LoRA Custom** reveals the nuanced impact of spatial transformations during fine-tuning. The introduction of stochastic rotation ($[-20^\circ, 20^\circ]$), cropping, and horizontal flipping acts as an effective regularizer, yielding a marginal improvement in text adherence (CLIP-T = 0.8723 vs. 0.8678). However, this regularization comes at the expense of subject fidelity (DINO-I = 0.6252 vs. 0.6303). We believe that, because subject-driven personalization requires preserving micro-level details, spatial transformations - particularly random cropping and rotation - slightly distort high-frequency pixel alignments. As a result, the unaugmented **LoRA Baseline** remains the superior approach for high-fidelity concept preservation.

Multi-Encoder Complexity in Textual Inversion. We believe that the poor performance of Textual Inversion stems primarily from the high architectural complexity of Stable Diffusion 3.5. Given its triple text-encoder system and joint MM-DiT backbone, optimizing continuous embeddings alone within a practical step count - and without fine-tuning the backbone’s internal text stream - is insufficient to steer the generation process effectively toward high-fidelity personalization.

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